$$\begin{aligned}\left\langle \hat{q}_i(\tau)\hat{q}_i \right\rangle_{\hat{\rho}\mathcal{E}} &= \frac{1}{2m_i\omega_i}\left\langle \hat{a}_i\hat{a}_i^\dagger \mathrm{e}^{-\mathrm{i}\omega_i\tau} + \hat{a}_i^\dagger\hat{a}_i \mathrm{e}^{\mathrm{i}\omega_i\tau} \right\rangle_{\hat{\rho}\mathcal{E}} \\ &= \frac{1}{2m_i\omega_i}\left\{ \left\langle \hat{a}_i\hat{a}_i^\dagger \right\rangle_{\hat{\rho}\mathcal{E}} \mathrm{e}^{-\mathrm{i}\omega_i\tau} + \left\langle \hat{a}_i^\dagger\hat{a}_i \right\rangle_{\hat{\rho}\mathcal{E}} \mathrm{e}^{\mathrm{i}\omega_i\tau} \right\}. \end{aligned} \tag{5.10}$$

But the quantity

$$N_i = \left\langle \hat{a}_i^\dagger\hat{a}_i \right\rangle_{\hat{\rho}\mathcal{E}} \tag{5.11}$$